SAI VARDHAN PALLERLA

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EDUCATION

VNR Vignana Jyothi Institute of Engineering & Technology — **Hyderabad, India**

Bachelor of Technology in Electronics and communication Engineering — *June 2019 – May 2023*

TECHNICAL SKILLS

Languages: C++, C, Python, Java, Perl, SQL, JavaScript
Systems Infrastructure: Linux, Docker, Kubernetes, Terraform, AWS, OpenStack, CI/CD
Networking & Protocols: TCP/IP, SNMP, IPSec, SSL/TLS, GRPC, REST APIs
Machine Learning & NLP: Lang Graph, Lang Chain, RAG Systems, Multi-Agent Workflows, Chroma DB, Fast API
Developer Tools & Frameworks: Git, Pytest, Playwright, Robot Framework, Postman, MySQL

EXPERIENCE

F5 Networks Inc — ***Aug 2023 – Present***

Software Engineer – II — *Hyderabad, India*

Automated AI Assistant for Software Testing & Bug Fixing

- Designed and engineered PRISM, an interactive **multi-agent AI system** that automates the end-to-end bug-fix lifecycle for BIG-IP network security defect orchestrating automated triage, source-level root cause analysis with confidence gating, and precise patch generation.
- Accelerated bug resolution timelines by **66%** (reducing defect lifecycle from 30 days to 10 days) across multiple cross-functional engineering teams, facilitating significantly faster release cadences while drastically reducing developer manual overhead.
- Designed and deployed an autonomous QA agent that parses feature requirement documents (PDRs), developer code, and REST APIs to generate end-to-end security, API, and UI test plans, slashing test planning cycles by **90%** (from 15 days to a single-day scan) across multiple F5 teams.
- Engineered a self-healing automation loop that generates and dry-runs executable test code, dynamically syncs test cases to the TestRail database, and autonomously orchestrates Git Merge Requests (MRs) upon developer validation, improving total test creation efficiency by **70%**.

IPsec VPN Support for the BIGIP

- Owned the technical delivery of the complex **IPsec VPN** user interface, designing both the intuitive frontend experience and the underlying backend data schemas to serve as the structural blueprint for all subsequent backend feature implementation.
- Engineered a seamless, automatic **fallback mechanism** from IPsec VPN to SSL-VPN on port 443 (HTTPS), ensuring uninterrupted network connectivity for enterprise users working behind restrictive guest firewalls or routers lacking NAT-T support.
- Designed and implemented EAP-MSChapv2 based authentication for IPSec VPN tunnel establishment on F5 BIG-IP APM, eliminating the need for per-client PKI certificate provisioning and reducing admin operational overhead.

Distributed Network Management & SNMP Support for BIGIP

- Architected containerized microservice deployments for network management daemons on Kubernetes-based BIG-IP Next pods, integrating CI/CD test gates to cut pipeline build execution times by **55%** and build failures by **45%.**
- Engineered high-throughput SNMP agent infrastructure for BIG-IP Next, authoring an automated Python/Perl code-generation engine that compiled 1,500+ MIB object handlers and configuration templates across containerized runtimes.

BIG-IP Deployment & SSL VPN Test Automation Framework

- Automated end-to-end provisioning of BIG-IP, client, and server environments on OpenStack using Terraform and Python, cutting setup time by **71% (3 hrs → 35 min)**, enabling rapid environment recreation for customer escalations and SRS.
- Built an AI-powered **Web Automation Agent** integrating Model Context Protocol (MCP) with Playwright, enabling autonomous, session-aware multi-step browser interactions across Chromium, Firefox, and WebKit for complex end-to-end UI test workflows.

Smart Interviews | Certificate — ***Sept 2022 – April 2023***

Teaching Assistant – (Data Structures & Algorithms, Full Stack Development) — *Hyderabad, India*

- Instructed **150+** undergraduate students in core Data Structures and Algorithms (DSA), conducting code reviews, space/time asymptotic complexity analysis, and algorithmic debugging in C++ and Java, Python.
- Mentored candidates through 1-on-1 technical interview coaching and problem-solving workshops, resulting in a **90%** placement rate into competitive SWE internship and full-time roles.

PROJECTS

Tailor CV | Chrome Extension — *JavaScript/TypeScript, Python, Fast API, Lang Graph, RAG, Chroma DB*

- Engineered a context-aware Chrome extension that enables students and working professionals to tailor resumes on the fly directly from job boards, boosting application-to-interview alignment and sustaining **90+** ATS compliance scores.
- Integrated client-side DOM scraping with an asynchronous backend powered by Fast API and Lang Graph, executing automated job-description parsing, vector retrieval via Chroma DB, and real-time prompt generation in sub-3 seconds.

ACHIEVEMENTS

- **1st** place, F5 Gemini Hackathon — built a Code Graph & Unified Memory based token optimization solution for LLM workloads, cutting cost and latency.
- Code Chef **4-Star** Coder: Ranked **#48** globally in a worldwide algorithmic coding challenge among thousands of international competitors.
- Elected Chairperson, **IEEE** Student Branch (ECE): Spearheaded technical initiatives for student members, driving a **40%** surge in active community participation and peer-led project development.